

VOCODE: A Systematic Evaluation of Quality, Efficiency, and Deployment Trade-offs in Neural Vocoders

Rahul Sawhney

RAHUL.SAWHNEY@FAU.DE

Department Artificial Intelligence in Biomedical Engineering

Friedrich-Alexander-Universität Erlangen-Nürnberg

Nürnberg Straße 74, 91052 Erlangen, Germany

Abstract

Neural vocoders differ sharply in quality, speed, and footprint, yet published results resist comparison: corpus, training budget, implementation, evaluator, and hardware vary across papers. VOCODE benchmarks twelve vocoders spanning autoregressive, GAN-based, iSTFT-based, flow-matching, and attention-augmented designs, trained from scratch on LJSpeech with architecture-native objectives and identical data splits. Evaluation is unified: ten quality metrics (PESQ, STOI, UTMOS, spectral and pitch distances) plus real-time factor, latency, parameters, and size. Deployment interventions yield 69 transformed groups paired with 22 same-hardware baselines (273 executions), each judged under an explicit decision rule: 19 beneficial, 32 harmful, 18 inconclusive. No vocoder is Pareto-optimal across quality, latency, and footprint: the speed-quality frontier is Vocos, RFWave, and RNDVoC. Gains were architecture-dependent: FP16 halved storage at preserved PESQ in eight of ten groups, quality-preserving speedups reached $2.07\times$ (ONNX) and $1.40\times$ (compilation), and RFWave at eight ODE steps gained $1.20\times$ within 0.05 PESQ. Masked pruning was harmful in 18 of 19; static INT8 compressed models near-fourfold at material quality loss. At moderate quality, parameter, and storage budgets, selection chose FP16-cast Vocos on GPU and dynamic-INT8 Vocos on CPU. With an adaptive evaluation set, unreplicated training, and no listening test, VOCODE supports conditional, auditable comparisons, not a causal architecture ranking.

Keywords: neural vocoder, speech synthesis, model efficiency, real-time factor, quantization, pruning, Pareto analysis, reproducibility

1 Introduction

A neural vocoder reconstructs speech from an acoustic representation. Autoregressive systems generate samples sequentially (van den Oord et al., 2016), whereas adversarial, inverse short-time Fourier transform (iSTFT), and flow models move learned computation toward parallel waveform synthesis, frame-level spectral prediction, or a few solver steps (Kong et al., 2020; Siuzdak, 2024; Liu et al., 2025). These computational graphs expose different bottlenecks, so parameter count, serial dependency, memory traffic, transform overhead, and solver evaluations need not track one another.

Published quality-efficiency rankings also mix corpora, preprocessing, training budgets, checkpoint rules, implementations, evaluators, and hardware. Even within one implementation, parameter count, serialized bytes, multiply-accumulate counts, and real-time factor (RTF) answer different questions. Deployment transformations widen the gap between a method name and the artifact that executes: compilation can break a graph, quantization

can cover only selected operators, and dense pruning can retain neither sparse storage nor sparse kernels (Han et al., 2015; Jacob et al., 2018; Ansel et al., 2024).

This work benchmarks twelve neural-vocoder configurations trained from scratch under identical data splits and one evaluation implementation. The cohort comprises APNet2, BigVGAN-base, FreeV, HiFi-GAN V1, V2, V3, LPCNet, MelGAN, RFWave, RNDVoC, Vocos, and VocosFormer. Architecture-native conditioning, objectives, and training budgets are retained. Each configuration contributes one training trajectory from recorded seed 1234 and one retained checkpoint. Inference seeds 42, 43, and 44 re-execute that frozen state rather than retrain; the single exception is static ONNX INT8, which rebuilds calibration state on each execution.

Three research questions structure the study. *RQ1*: under identical data splits and one evaluation implementation, but architecture-native objectives and budgets, how do the twelve configurations trade objective quality, inference latency, parameter count, and deployable storage? *RQ2*: relative to same-hardware baselines, how do executed deployment transformations shift objective quality, warm real-time factor, and serialized artifact size, and how strongly do those effects depend on architecture and execution backend? *RQ3*: which measured deployment artifact is selected under explicit hardware-specific constraints on quality, parameter count, and deployable storage? The three are dependent analyses of one experiment: transformed variants inherit the trained checkpoints, selection consumes their measurements, and unsupported transformations are excluded on record.

The scope is deliberately conditional. LJSpeech contains one speaker. The final 525 utterances form an adaptive evaluation set: their PESQ values informed checkpoint decisions and one deterministic repair of RFWave’s inference sampler. Objective measures are proxies, not listener judgments (Rix et al., 2001; Taal et al., 2011; Saeki et al., 2022). Population-level or causal claims are precluded by one training trajectory per configuration, unequal architecture-native budgets, independently scheduled training jobs, and the absence of listening or multi-speaker experiments.

2 Literature Review

The literature separates into generation mechanisms, deployment transformations, and comparative evaluation; the unqualified label *vocoder efficiency* conflates model structure, executable representation, and measurement protocol.

2.1 Generation Mechanisms

WaveNet factorizes waveform likelihood sample by sample (van den Oord et al., 2016); LPCNet reduces learned synthesis through linear prediction but retains a recurrent excitation generator (Valin and Skoglund, 2019). Adversarial vocoders instead use parallel generators. MelGAN combines multi-scale discrimination with feature matching (Kumar et al., 2019); HiFi-GAN adds multi-period discrimination and quality-footprint variants (Kong et al., 2020); BigVGAN introduces periodic activations and anti-aliased representations (Lee et al., 2023). These mechanisms differ in critical-path parallelism, so capacity alone cannot predict latency.

iSTFT vocoders perform learned prediction near the acoustic-frame rate and delegate waveform reconstruction to overlap-add; iSTFTNet established the pattern by replacing a

time-domain generator’s final upsampling stages with the inverse transform (Kaneko et al., 2022). APNet2 predicts amplitude and phase spectra (Du et al., 2023); Vocos predicts complex coefficients with a constant-resolution ConvNeXt stack (Siuzdak, 2024); FreeV uses a pseudo-inverse mel prior (Lv et al., 2024); and RNDVoC separates analytical range and learned null-space components (Li et al., 2025). RFWave instead integrates a rectified flow in a multi-band spectral representation (Liu et al., 2025), making solver evaluations an explicit latency-quality control.

VocosFormer is not a published architecture: the project inserts residual-convolution and self-attention components into a capacity-matched Vocos chassis, adopting them from the WavTokenizer decoder lineage (Ji et al., 2025); the codec is cited for component provenance only.

2.2 Deployment Transformations

Pruning changes both numerical weights and, only when supported, their storage and execution representation. Classical train-prune-retrain work couples sparsity to compressed storage (Han et al., 2015); dense zero masks establish tolerance to weight removal but not acceleration. Quantization likewise specifies weights, activations, scales, zero-points, covered operators, and backend (Jacob et al., 2018). CPU dynamic quantization, ONNX QDQ graphs, and GPU weight-only quantization are therefore distinct treatments despite sharing an integer bit width. Compilation changes the captured executable graph; graph breaks, specialization mode, operator support, and hardware determine whether TorchDynamo/TorchInductor produces a gain (Ansel et al., 2024). The vocoder literature itself pursues efficiency through architecture and training: WaveRNN prunes its recurrent weights during training to reach real-time mobile-CPU synthesis (Kalchbrenner et al., 2018), and WaveFM distills a flow-matching vocoder toward single-step synthesis (Luo et al., 2025); the transformations above instead operate on fixed trained checkpoints. Reproducible deployment evidence must therefore state transformation scope, backend, operator coverage, and the paired pre-transformation measurement, not transformation names alone.

2.3 Comparative Evaluation

PESQ models perceived speech distortion (Rix et al., 2001), STOI estimates intelligibility (Taal et al., 2011), and UTMOS predicts mean opinion scores (Saeki et al., 2022); agreement among these proxies is still not a listening test. Shared-protocol vocoder comparisons improve internal validity while remaining sensitive to implementation and conditioning choices (Govalkar et al., 2019), and VocBench trains and evaluates six vocoders in one shared environment with objective and subjective scoring (AlBadawy et al., 2022). Deployment-focused work shows that batch speed does not establish low-latency streaming performance because causality, chunk size, lookahead, loading overhead, and hardware can reverse rankings (Yoneyama et al., 2025). VOCODE does not claim streaming evaluation or listener scoring. Relative to VocBench it contributes twelve configurations trained from scratch under recorded architecture-specific budgets, explicit checkpoint provenance, inclusion and exclusion registers for executable deployment artifacts, paired before-and-after measurements on one hardware profile, and constraint-aware selection.

3 Methodology

We benchmark twelve neural vocoders under one common data, evaluation, and measurement protocol. A *Project-Trained Configuration* is one architecture recipe fitted from random initialization under the project budget; its selected state is the *Retained Project Checkpoint*, which every measurement inherits. The protocol fixes corpus membership, evaluator, and measurement procedure while each configuration keeps its native conditioning, objective, and optimization recipe: a structured common testbed, not an equal-compute causal benchmark. Throughout, a deployment artifact $a = (m, t, h)$ applies transformation t to the checkpoint θ_m^* of configuration $m \in \mathcal{M}$ on requested *hardware lane* $h \in \{\text{B200, CPU}\}$; the identity transformation defines the paired same-lane control $c(a)$, and inference seeds $s \in \{42, 43, 44\}$ re-execute one selected state without resampling training (single exception: static ONNX INT8, Section 3.4).

3.1 Data Set and Preprocessing

All experiments use LJSpeech 1.1, 13,100 English utterances from one speaker (Ito and Johnson, 2017). Identifier-sorted disjoint blocks fix 12,475 training, 100 validation, and 525 evaluation items without a random seed; identifier order preserves chapter structure, so the split is fixed membership rather than a random utterance sample. Training loaders shuffle and crop fixed-length segments, whereas validation and evaluation consume complete utterances. The evaluation suffix is *adaptive*, not untouched: its PESQ values informed stop and extension decisions, and one deterministic repair of RFWave’s inference sampler.

Every configuration retains its native conditioning interface: most consume 80 Slaney-scale mel bands at 22.05 kHz; BigVGAN-base uses 100 normalized bands, and Vocos, VocosFormer, and RFWave use 100 unnormalized HTK-scale bands at 24 kHz; all mel pipelines share a 1,024-point Hann transform with a 256-sample hop, while LPCNet consumes 18 Bark-frequency cepstra plus pitch and correlation features at 16 kHz. Training-only transformations are peak normalization to 0.95 (HiFi-GAN V1, V2, and V3), uniform random peak gain in $[-6, -1]$ dB (Vocos, VocosFormer), and LPCNet excitation noise with standard deviation 0.3; evaluation waveforms receive none.

3.2 Model Architectures

The cohort spans five generation mechanisms: time-domain adversarial upsamplers (MelGAN, HiFi-GAN V1, V2, V3, BigVGAN-base), adversarially trained spectral models with inverse-STFT heads (APNet2, FreeV, RNDVoC, Vocos), hybrid attention (VocosFormer), sample-level autoregression (LPCNet), and rectified-flow synthesis (RFWave); Table 1 reports the executed configuration surface, with the cited architectural definitions in the literature review.

VocosFormer, the project-defined exception, inserts a residual-convolution, self-attention, residual-convolution block from the WavTokenizer decoder lineage into a shallower Vocos chassis under matched losses, data, and training quantum; its 13.78 million parameters sit 1.83% above Vocos, and one training trajectory supports only configuration-level comparison.

Configuration	Family	Conditioning	Params	Segment	Batch	Prec.	Updates	State
APNet2	iSTFT GAN	80 mel / 22.05	31.43	8,192	32	BF16	124.8	120k
BigVGAN-base	Waveform GAN	100 mel / 24	14.03	8,192	32	BF16	124.8	120k
FreeV	iSTFT GAN	80 mel / 22.05	18.22	8,192	32	BF16	124.8	120k
HiFi-GAN V1	Waveform GAN	80 mel / 22.05	13.94	8,192	32	BF16	254.8	245k
HiFi-GAN V2	Waveform GAN	80 mel / 22.05	0.93	8,192	32	BF16	124.8	120k
HiFi-GAN V3	Waveform GAN	80 mel / 22.05	1.46	8,192	32	BF16	254.8	245k
LPCNet	Autoregressive	20 feat / 16	1.23	2,400	128	FP32	31.3	30k
MelGAN	Waveform GAN	80 mel / 22.05	4.27	16,000	32	BF16	124.8	120k
RNDVoC	iSTFT GAN	80 mel / 22.05	3.57	16,384	32	BF16	124.8	120k
RFWave	Rectified flow	100 mel / 24	18.14	32,512	64	BF16	125.0	120k
Vocos	iSTFT GAN	100 mel / 24	13.53	16,384	32	BF16	124.8	120k
VocosFormer	iSTFT GAN + attn	100 mel / 24	13.78	16,384	32	BF16	124.8	120k

Table 1: Project-Trained Configurations. Conditioning: input features and rate (kHz); Params: deployable parameters (millions); Segment: training crop (samples); Updates: executed budget (thousands); State: the Retained Project Checkpoint, the terminal durable state selected by the registered gate.

3.3 Training and Checkpoint Selection

We fitted each configuration through one recorded-seed (1234) training trajectory under its native objective and a finite registered budget. All models are PyTorch modules executed through the project-controlled SyntheticMind harness, which standardizes fitting, validation, metric reduction, and checkpoint restore; the VOCODE library supplies data pipelines, objectives, synthesis paths, evaluators, and deployment transformations, and each configuration, variant, lane, and execution runs as one isolated job. Eleven runs used a 48-GB NVIDIA L40S and RFWave an 80-GB NVIDIA A100; recorded training sums to 75.31 GPU-hours, a lower bound. Architecture-native rather than equalized budgets make the retained states controlled project realizations, not attainable quality ceilings.

Objectives are architecture-native. MelGAN optimizes least-squares adversarial and feature-matching terms alone; HiFi-GAN V1, V2, and V3 and BigVGAN-base combine the same terms over their discriminator ensembles with weighted mel reconstruction, and Vocos and VocosFormer apply the hinge form of that composition. APNet2, FreeV, and RNDVoC add log-amplitude, anti-wrapping phase, and STFT-consistency terms with real and imaginary reconstruction to mel, hinge adversarial, and feature-matching supervision. LPCNet minimizes teacher-forced cross entropy over its quantized mu-law excitation; RFWave regresses conditional flow velocities in the waveform domain with auxiliary magnitude and band-overlap terms. Objectives were deliberately not standardized, so raw training and validation losses are not comparable across families and never ranked configurations. Ten configurations optimized with AdamW and two (MelGAN, LPCNet) with Adam; initial learning rates were 2×10^{-4} (APNet2, FreeV, HiFi-GAN V1, V2, and V3, RNDVoC, RFWave), 1×10^{-4} (BigVGAN-base, MelGAN), 5×10^{-4} (Vocos, VocosFormer), and 1×10^{-3} (LPCNet). LPCNet additionally trains under scheduled GRU sparsification from 2k to 20k updates, the

HiFi-GAN V1 and V3 budgets include 4,800 replayed updates after a durable resume, and exact coefficients and schedules remain recoverable from the resolved run records and code.

Checkpoint selection follows a registered gate: validation loss informed continuation, durable checkpoints were written every 5,000 updates, a sub-0.02 PESQ gain across gates indicated a plateau, and every run kept a registered ceiling. Where a commensurate published or released-checkpoint PESQ anchor existed, the registered quality target was 93% of that anchor; RFWave used a registered PESQ floor of 3.55, and LPCNet and VocosFormer had no commensurate numeric anchor. For each configuration, evaluation used the terminal durable checkpoint admitted by this gate, the Retained Project Checkpoint θ_m^* of Table 1, not a retrospectively substituted validation minimum; the validation-best, global-minimum, and evaluated states remain in the resolved records. The accepted runs predate the repair that seeds model construction and workers deterministically, so 1234 is a recorded rather than reconstructible seed; retained states are immutable, and bitwise regeneration is not claimed.

3.4 Deployment Optimization Methods

We transform each Retained Project Checkpoint with the deployment variants of Figure 1, spanning TorchInductor compilation, FP16 weight casting, INT8 quantization, ONNX deployment, magnitude pruning at 30/50/70% with one recovery arm, and RFWave solver-step reduction. Admission requires an executable transformed path, nonzero operator or parameter coverage, all 525 utterances under three executions, and a same-profile pre-transformation measurement; unsupported cells fail before execution expansion. The executed surface contains 22 pre-transformation controls and 69 transformed groups (91 groups, 273 executions); LPCNet completed no admissible execution on either lane, leaving eleven transformed configurations, and 29 exclusion records document 88 rejected attempts.

Static ONNX INT8 uses ONNX Runtime 1.24.4 on its CPU provider with eight intra-op threads, QDQ format, per-channel QInt8 weights, QUInt8 activations, and coverage of Conv, ConvTranspose, MatMul, and Gemm. Each execution reconstructs its quantized graph, calibrating on 48 mel batches from the execution-seeded shuffled training loader, so these executions combine calibration-set and runtime variation rather than repeating one fixed artifact. Dynamic and weight-only INT8 retain their distinct PyTorch and torchao semantics; an unsupported INT4 probe contributes no result. Pruning is magnitude-based with dense masks, testing weight removal but not sparse storage or kernels: Vocos, FreeV, VocosFormer, and RFWave use L_2 -ranked structured output-channel pruning over backbone Linear projections (spectral heads excluded); the rest use global L_1 unstructured pruning over convolutional, Linear, and recurrent weights. Selected 50% states received 13 recovery epochs at one tenth learning rate, masks held during fine-tuning and zeros baked into the evaluated checkpoint; absent an admitted dense-continuation control, recovered rows support descriptive rather than causal claims. RFWave step reduction shrinks its ten-step Euler baseline to eight, four, or two steps.

3.5 Evaluation Metrics and Analysis Protocols

Quality proxies. Wideband PESQ, withdrawn by ITU-T in January 2024 in favor of the P.863 family (ITU-T, 2024) yet still the dominant quality proxy in the vocoder literature,



Figure 1: Deployment admission surface. Filled: admitted three-execution groups (91, incl. 22 same-lane controls); crosses: the ten excluded attempts inside the plotted vocabulary (invalid recovery or control states; incomplete LPCNet executions), with the 29-row exclusion register remaining authoritative. Compile RO = reduce-overhead; INT8 WO = weight-only; Recovery = 13-epoch recovery of the 50%-pruned state; ODE k = k Euler steps.

remains the primary intrusive measure (Rix et al., 2001); STOI (Taal et al., 2011) and UTMOS (Saeki et al., 2022) test dependence on the proxy choice. Reference and synthesized waveforms are truncated to their common unpadded length (*true-length* evaluation) before resampling; PESQ operates wideband at 16 kHz, classic non-extended STOI at 10 kHz, and UTMOS (SpeechMOS UTMOS22 strong 1.2.0) scores the synthesized waveform alone at 16 kHz. Mel L_1 , multi-resolution STFT, mel-cepstral, log-spectrum, pitch, periodicity, and voicing errors are diagnostics only; architecture-native or alignment-sensitive values are not cross-model endpoints, and without a blinded listening study every quality claim remains a proxy claim. For utterance i in the fixed 525-item evaluation set $\mathcal{I}$, with conditioning x_i and reference y_i , artifact a under execution s synthesizes $\hat{y}_{a,s,i} = f_{a,s}(x_i)$, and proxy q is aggregated as

$$\hat{Q}_a^{(q)} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} q(y_i, \hat{y}_{a,s,i}), \quad \mathcal{S} = \{42, 43, 44\}, \quad (1)$$

where the reference-free UTMOS ignores y_i ; execution values are averaged within each utterance before any paired contrast, and for static ONNX INT8 the execution index also carries calibration-state variation (Section 3.4).

Timing and footprint protocol. Beyond quality, endpoints include the continuous process-RSS high-water mark, one-time ONNX session-construction time, and complexity as parameters, serialized bytes, and MACs from one traced second of audio; jobs were independently scheduled, and paired contrasts match the requested resource profile, never the physical host. After five warm-up batches, pre-transformation evaluation measures one complete pass at batch one on an NVIDIA B200. The CPU column of Table 2 is the retained

comparative record, three executions each timing one complete post-warm-up pass on an 8-core CPU host, and is never cross-compared with deployment-phase RTFs; LPCNet’s 27-item timing record is excluded from strict speed frontiers. Deployment evaluation measures three synchronized calls per utterance for the 520 post-warm-up utterances on B200 or CPU, each timed call following one untimed prediction of the same input, so the quantities describe warm repeated synthesis only. For artifact a , execution s , timed items $\mathcal{I}_a \subseteq \mathcal{I}$, synchronized time $\tau_{a,s,i,r}$, duration d_i , and $J_a \in \{1, 3\}$ calls, warm RTF is

$$\bar{\tau}_{a,s,i} = J_a^{-1} \sum_{r=1}^{J_a} \tau_{a,s,i,r}, \quad R_{a,s} = \frac{\sum_{i \in \mathcal{I}_a} \bar{\tau}_{a,s,i}}{\sum_{i \in \mathcal{I}_a} d_i}, \quad (2)$$

and the group-level $\hat{R}_a$ is the mean of $R_{a,s}$ over executions. Reported p50/p95 are per-execution percentiles of the 520 per-utterance three-repetition means $\bar{\tau}_{a,s,i}$, averaged over executions, not of the 1,560 individual calls, whose raw durations were not retained. B200 jobs requested 16 CPUs and 64 GiB, CPU jobs 8 CPUs and 16 GiB; the Python 3.14.2, PyTorch 2.13.0+cu130, torchaudio 2.11.0, torchao 0.17.0, ONNX 1.20.1, and ONNX Runtime 1.24.4 stack is retained, whereas host-identifying fields are irrecoverable.

Paired contrasts and decision rule. For quality proxy q and transformed artifact a with paired control $c(a)$, quality estimates $\hat{Q}$ from Eq. (1), warm RTF $\hat{R}$ from Eq. (2), and serialized deployable size D , the continuous effects are

$$\Delta Q_a^{(q)} = \hat{Q}_a^{(q)} - \hat{Q}_{c(a)}^{(q)}, \quad S_a = \hat{R}_{c(a)} / \hat{R}_a, \quad Z_a = D_a / D_{c(a)}. \quad (3)$$

Quality deltas are utterance-paired, whereas speedups and size ratios are same-profile descriptive contrasts because hosts are not paired. Quality contrasts pair the execution-averaged utterance values of Eq. (1) before a Student t interval, with a 5,000-replicate percentile bootstrap as sensitivity analysis (Student, 1908; Efron, 1979). These are fixed-set descriptive intervals quantifying utterance dispersion over the 525 fixed items under a working exchangeability assumption; they omit multiplicity correction and variation from training, speaker, recovery seed, or host, and for static INT8 do not separate calibration from runtime variation. A post hoc ordered classifier provides an exploratory engineering summary of Eq. (3): dense-masked pruning and recovery rows are labeled first: harmful when $\Delta Q_a^{(\text{PESQ})} < -0.10$, otherwise inconclusive; remaining MelGAN rows are fixed as inconclusive because their PESQ is floor-censored. Every other row receives

$$\ell(a) = \begin{cases} \text{harmful,} & \Delta Q_a^{(\text{PESQ})} < -0.10 \vee (S_a < 0.90 \wedge Z_a > 0.90) \\ \text{beneficial,} & \Delta Q_a^{(\text{PESQ})} \geq -0.05 \wedge S_a \geq 0.90 \wedge (S_a \geq 1.10 \vee Z_a \leq 0.55) \\ \text{inconclusive,} & \text{otherwise.} \end{cases} \quad (4)$$

Thresholding PESQ alone makes these labels PESQ-preserving statements, not perceptual equivalence claims. A $3^5 = 243$ grid tests label sensitivity by varying each cutoff in Eq. (4) over three levels (preservation $-0.075 / -0.05 / -0.025$; harm $-0.15 / -0.10 / -0.075$; speed floor $0.85 / 0.90 / 0.95$; gain $1.05 / 1.10 / 1.20$; size $0.50 / 0.55 / 0.60$).

Frontier and constraint-aware selection. Under one quality proxy q and one cost axis (RTF, parameter count, or serialized size), artifact a dominates a' when $\widehat{Q}_a^{(q)} \geq \widehat{Q}_{a'}^{(q)}$ at no greater cost, with at least one inequality strict; each reported frontier is the nondominated set, recomputed per proxy and axis. For hardware lane h , admitted artifacts $\mathcal{A}_h$, quality floor $q_{\min}$, parameter ceiling $P_{\max}$, and size ceiling $D_{\max}$, deployment selection solves

$$a_h^* \in \arg \min_{a \in \mathcal{A}_h} \widehat{R}_a \quad \text{s.t.} \quad \widehat{Q}_a^{(q)} \geq q_{\min}, \quad P_a \leq P_{\max}, \quad D_a \leq D_{\max}, \quad (5)$$

so every recommendation is conditional on hardware, proxy, and constraints.

4 Results

Results cover the twelve Project-Trained Configurations and the 91 admitted model-variant groups of Figure 1: 22 same-lane controls and 69 transformed groups, measured over 273 executions. All quality values are the estimator of Eq. (1) on the fixed adaptive LJSpeech evaluation set, and every claim concerns these executed states. Except for static ONNX INT8 (Section 3.4), the three executions per group repeat inference from one frozen artifact, so their spread reflects execution variability rather than statistical uncertainty.

4.1 Baseline Quality, Efficiency, and Footprint

Addressing RQ1, Table 2 reports the complete twelve-configuration outcome surface, including the floor-level negative results, and Figure 2 places every configuration on the quality-speed plane. True-length PESQ ranked RNDVoC (3.895), RFWave (3.785), and Vocos (3.614) as the three strongest states, and STOI produced the same leading order; seven configurations reached true-length PESQ of at least 3.30. UTMOS disagreed with the intrusive proxies: HiFi-GAN V1 was highest at 4.203, followed by Vocos (4.089) and BigVGAN-base (4.062); no configuration was uniformly best across the three proxies.

No checkpoint minimized all deployment costs. Vocos had the lowest strict B200 RTF (0.00473), RNDVoC attained the highest PESQ at 11.8 times that RTF, and RFWave occupied the intermediate point (3.785 at 0.0148); the strict speed frontier comprised Vocos, RFWave, and RNDVoC. HiFi-GAN V2 was the smallest baseline (0.93 million parameters, 3.54 MB) at PESQ 2.400; the parameter and size frontiers comprised HiFi-GAN V2, HiFi-GAN V3, and RNDVoC. Two configurations produced floor-level negative results: MelGAN (PESQ 1.043 on both lanes) is consistent with budget-limited non-convergence, and LPCNet (1.052) lost free-running stability at 5.8% of its reference recipe while its PyTorch timing omits the author’s optimized sparse C path; both remain full members of the twelve-model cohort.

The matched VocosFormer state did not improve on its Vocos control: it added 247,040 parameters (+1.83%), reduced true-length PESQ by 0.101 and STOI by 0.0035, and increased B200 RTF 2.58 $\times$. Under padded batch-16 evaluation the same checkpoint measured PESQ 2.480 against 3.512 at true length, a padding-sensitivity failure consistent with contamination through unmasked global attention rather than intrinsic synthesis collapse; this one-trajectory comparison is a configuration-level observation, not an average attention effect.

Configuration	Params	Size	PESQ	STOI	UTMOS	RTF B200	RTF CPU [§]
APNet2	31.43	119.88	2.515	0.930	3.091	0.0095	0.0191
BigVGAN-base	14.03	53.51	3.320	0.974	4.062	0.1573	0.6985
FreeV	18.22	69.66	3.386	0.969	3.789	0.0077	0.0157
HiFi-GAN V1	13.94	53.16	3.305	0.974	4.203	0.1197	0.1836
HiFi-GAN V2 [‡]	0.93	3.54	2.400	0.942	3.907	0.1140	0.0507
HiFi-GAN V3 [‡]	1.46	5.59	2.550	0.949	3.851	0.0580	0.0366
LPCNet	1.23	4.70	1.052	0.527	1.235	11.36*	9.2015
MelGAN	4.27	16.27	1.043	0.418	1.736	0.0742	0.0456
RFWave [†]	18.14	69.21	3.785	0.981	3.514	0.0148	0.7491
RNDVoC ^{†‡}	3.57	14.94	3.895	0.987	4.007	0.0559	0.0838
Vocos [†]	13.53	51.62	3.614	0.979	4.089	0.0047	0.0098
VocosFormer	13.78	52.57	3.512	0.976	3.901	0.0122	0.0105

Table 2: Baseline outcome surface for all twelve configurations: quality proxies (higher is better), parameters (millions), serialized size (MB), and warm real-time factors (lower is better). [§]retained comparative CPU record (Section 3.5), not a deployment-phase value; [†]B200 speed-quality frontier; [‡]parameter- and size-quality frontier; *LPCNet timing (27 utterances, one execution) is excluded from strict frontiers.

4.2 Deployment Transformation Effects

Addressing RQ2, the decision rule of Eq. (4) labeled the 69 transformed groups 19 beneficial, 18 inconclusive, and 32 harmful. Table 3 reports the seven intervention families and their fourteen variants with per-variant label composition, median same-lane effects, and Δ PESQ ranges; the 29-row exclusion register documents the 88 attempts producing no admitted group. FP16 weight casting was the most consistently favorable family: eight of ten groups were beneficial at half serialized size, with RFWave and HiFi-GAN V3 reaching $1.955\times$ and $1.874\times$ speedups at preserved PESQ.

Compilation was mode specific: default-mode compilation preserved PESQ in all eleven groups at speedups from $0.14\times$ to $1.403\times$ (HiFi-GAN V3), whereas the reduce-overhead mode was harmful for three of four groups (speedups 0.34 to 0.41) and beneficial only for RFWave ($1.361\times$). ONNX FP32 accelerated HiFi-GAN V2 by $2.071\times$ at Δ PESQ -0.00013 (fixed-set 95% interval $[-0.00036, 0.00010]$) yet split 1/2/1 (B/I/H) overall. Dynamic INT8 diverged: beneficial for APNet2 and Vocos, harmful for FreeV (-0.123) and RFWave (-0.648). Weight-only INT8 preserved PESQ in all four groups but slowed FreeV ($0.851\times$) and VocosFormer ($0.617\times$).

Static ONNX INT8 was harmful in three of four groups with median PESQ loss 0.770 at median size ratio 0.265. Because each execution recalibrates on its own shuffled loader batches, its per-execution PESQ disperses where all other variants repeat exactly: HiFi-GAN V1 measured 2.003, 2.053, and 1.910 against its execution-invariant ONNX FP32 value of 3.317. Dense magnitude masks were harmful in 4/4, 10/11, and 4/4 groups at 30/50/70% sparsity (median Δ PESQ -1.03 , -1.06 , and -2.27); the single non-harmful group is floor-censored MelGAN. Recovery returned four of five admitted 50% groups to within 0.05 PESQ of their controls (BigVGAN-base moved from -1.149 to $+0.017$) but left

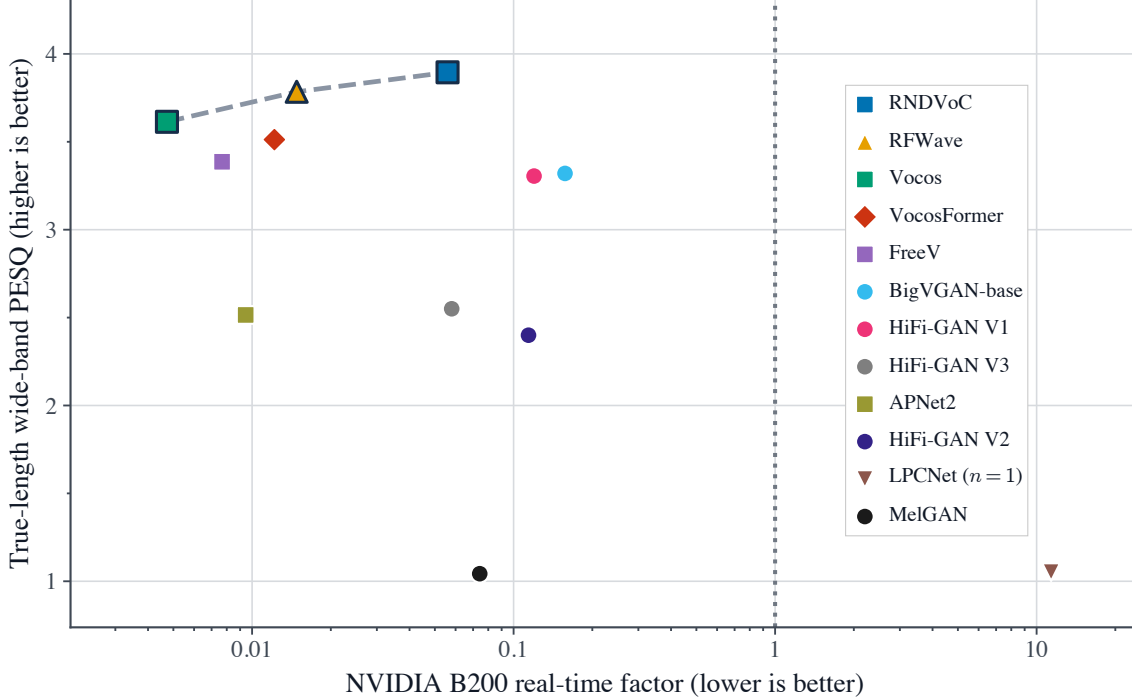


Figure 2: Twelve frozen baselines on the true-length PESQ versus B200 real-time-factor plane (Eq. (2)). The legend identifies each configuration, ordered by PESQ; marker shape encodes the generation family (squares: frame-level Fourier; circles: time-domain GAN; triangle: flow matching; diamond: hybrid attention; inverted triangle: autoregressive). Dark-edged markers are frontier members, connected by the dashed nondominance chain; the dotted line marks real-time synthesis. LPCNet’s single-execution timing is shown but excluded from the strict frontier. Each point is one row of the retained comparative record.

Vocos harmful (-0.475); lacking the dense-continuation control, the recovery arm supports no causal claim. RFWave solver reduction traded monotonically: speedups of 1.201, 2.200, and 4.509 for PESQ losses of 0.046, 0.369, and 1.636 at 8, 4, and 2 steps.

Deployment-phase RTFs follow Eq. (2) with $J_a=3$, not comparable with Table 2’s single-pass B200 or comparative CPU baselines (Section 3.5). Across the 273 executions, p50 spanned 1.21 to 3,909.35 ms, p95 1.96 to 6,138.02 ms, process RSS 6,754 to 16,991 MiB, and the 24 ONNX session builds spanned 0.0116 to 0.1327 s. The deployment costs need not move together: the HiFi-GAN V2 ONNX FP32 artifact halves median synthesis time on CPU while raising process RSS above its control. Of the 69 utterance-paired PESQ contrasts, 34 fixed-set intervals were entirely negative, two entirely positive, and 33 crossed zero. Despite non-normality (113 contrasts with absolute skewness above one, 181 with excess kurtosis above three), bootstrap and Student t signs agreed on 482 of 483 contrasts, an internal agreement diagnostic, not external validity.

Variant	n	B/I/H	Med. Δ PESQ	Δ PESQ range	Med. S_a	Med. Z_a
<i>Compilation</i>						
Default mode	11	4/5/2	-0.0000	$[-0.000, +0.000]$	1.07	1.00
Reduce-overhead mode	4	1/0/3	-0.0000	$[-0.000, +0.000]$	0.40	1.00
<i>FP16</i>						
Weight cast	10	8/2/0	-0.0002	$[-0.002, +0.001]$	1.08	0.50
<i>INT8</i>						
Dynamic	5	2/1/2	-0.0469	$[-0.648, +0.004]$	1.30	0.40
Weight-only	4	2/2/0	-0.0010	$[-0.003, +0.002]$	0.92	0.41
<i>ONNX</i>						
FP32 export	4	1/2/1	-0.0000	$[-0.000, +0.000]$	0.90	1.00
INT8 static	4	0/1/3	-0.7701	$[-1.329, -0.004]$	0.82	0.26
<i>Pruning</i>						
30% sparsity	4	0/0/4	-1.0259	$[-2.740, -0.148]$	0.95	1.00
50% sparsity	11	0/1/10	-1.0632	$[-2.719, +0.000]$	1.00	1.00
70% sparsity	4	0/0/4	-2.2722	$[-2.712, -2.014]$	1.03	1.00
<i>Recovery</i>						
50% + recovery	5	0/4/1	-0.0070	$[-0.475, +0.017]$	1.11	1.00
<i>ODE steps</i>						
8 steps	1	1/0/0	-0.0456	n/a	1.20	1.00
4 steps	1	0/0/1	-0.3689	n/a	2.20	1.00
2 steps	1	0/0/1	-1.6363	n/a	4.51	1.00

Table 3: Deployment intervention outcomes over the 69 admitted groups, by family and variant: admitted count, beneficial/inconclusive/harmful composition under Eq. (4), median same-lane Δ PESQ, speedup S_a , and size ratio Z_a (Eq. (3)), and the Δ PESQ range across the variant’s admitted groups; single-group variants report their one contrast.

4.3 Sensitivity of the Decision Taxonomy and Constraint-Aware Selection

Across the $3^5 = 243$ threshold profiles of Section 3.5, the beneficial, harmful, and inconclusive totals ranged over 12 to 24, 29 to 34, and 12 to 27; median and minimum agreement with the default taxonomy were 91.3% and 82.6%. Stability stratifies by rule structure: 27 of the 28 rows governed by the dense-mask and MelGAN overrides never changed label (96.4%), whereas 25 of the 41 threshold-governed rows never changed (61.0%), so the aggregate stability (52 of 69 rows unchanged) partly reflects deterministic rule structure rather than threshold insensitivity.

Frontier membership depended on the proxy: STOI reproduced the PESQ frontiers of Section 4.1, while UTMOS yielded Vocos then HiFi-GAN V1 against B200 RTF and, against parameters, HiFi-GAN V2, RNDVoC, Vocos, and HiFi-GAN V1, so proxy choice alters which states appear efficient.

Answering RQ3, Table 4 instantiates the selector of Eq. (5) over all admitted artifacts of each lane, using deployment-phase quality, RTF, deployable size, and logical parameter counts consistently. Under the moderate operating point ($q_{\min}=3.30$, $P_{\max}=20\text{M}$, $D_{\max}=70\text{ MB}$), 28 B200 artifacts and eight CPU artifacts remain feasible, and the selector chooses FP16-cast

Constraints ($q_{\min}, P_{\max}, D_{\max}$)	Lane	Feasible	Selected artifact a_h^*	RTF	PESQ
(3.30, 20M, 70 MB)	B200	28	FP16-cast Vocos	0.0003	3.611
(3.30, 20M, 70 MB)	CPU	8	Dynamic-INT8 Vocos	0.0055	3.563
(3.60, 15M, 55 MB)	B200	10	FP16-cast Vocos	0.0003	3.611
(3.60, 15M, 55 MB)	CPU	2	Untransformed Vocos	0.0067	3.610

Table 4: Constraint-aware selection (Eq. (5)) over all admitted artifacts per lane, with same-lane controls: feasible-set size and RTF-minimizing artifact per constraint vector (deployment-phase PESQ floor, parameter ceiling, size ceiling). RTF values follow the deployment-phase three-repetition estimand (not comparable with Table 2).

Vocos on B200 and dynamic-INT8 Vocos on CPU; tightening to (3.60, 15M, 55 MB) leaves ten and two feasible artifacts, selecting FP16-cast Vocos on B200 and the untransformed Vocos control on CPU. The feasible set remains nonempty on the B200 lane for quality floors up to RNDVoC’s FP16 artifact at 3.894; above that value no admitted artifact qualifies.

5 Discussion

RQ1 has no scalar answer. RNDVoC’s range-null decomposition produced the strongest intrusive-proxy state at modest parameter count, Vocos converted frame-level spectral prediction into the lowest RTF on both lanes, and RFWave traded iterative solver work for an intermediate point; the small HiFi-GAN variants minimized footprint below the leading quality band. Generation granularity and analytical output structure tracked the observed orderings better than parameter count alone, a mechanism-consistent reading, not causal identification, under native budgets and single trajectories. The choice of quality proxy is itself a deployment decision: PESQ and STOI selected RNDVoC while UTMOS selected HiFi-GAN V1 and reshaped both cost frontiers; proxy agreement still would not constitute listener preference. RQ3 inherited this conditionality: the selected artifact changed with lane and constraint vector.

RQ2 resolved conditionally on operator graph, backend, and runtime. FP16 casting halved storage almost uniformly, consistent with its arithmetic-format origin (Micikevicius et al., 2018), yet its latency effect spanned $0.554\times$ to $1.955\times$; compilation and ONNX gains depended on graph lowering and kernel choice (Ansel et al., 2024); and INT8 coverage was necessary but insufficient: transformed paths could shrink bytes yet regress speed (Jacob et al., 2018). Static ONNX INT8 also exposed calibration dependence: execution-seeded recalibration varied the quantized artifact by 0.14 PESQ for HiFi-GAN V1; deployed integer pipelines should treat the calibration set as part of the artifact’s identity. With hosts unpaired, large contrasts are more credible than marginal crossings, and the taxonomy remains an exploratory summary of continuous effects.

Dense masks and solver reduction manipulated different objects. Masks measured tolerance to zeroed weights, not sparse deployment (Han et al., 2015); recovery showed quality is relearnable, BigVGAN-base regaining 1.17 PESQ, but the missing dense-continuation control confounds recovery with added training. RFWave’s step ladder changed the iteration

count and produced the matrix’s only smooth, monotone speed-quality curve, making eight steps a defensible choice. VocosFormer supports a bounded mechanism hypothesis: unmasked global attention can spread padded-frame contamination, matching the padded-to-true-length gap; true-length evaluation removed most of the quality gap, none of the latency penalty, and attribution requires masked- and local-attention controls.

In this cohort, post hoc compression split sharply: FP16 held PESQ at half size in eight of ten groups, while dense masking harmed 18 of 19 groups and static ONNX INT8 three of four; the larger efficiency separations were architectural, consistent with the architecture-first emphasis of recent designs (Siuzdak, 2024; Liu et al., 2025; Lee et al., 2025; Oh et al., 2026).

All intervals are fixed-set descriptive quantities conditional on the adaptive single-speaker suffix, not an untouched test set; no multiplicity correction, training replication, listening study, or external evaluation exists, and warm timing omits streaming, queuing, cold start, and host identity. The priority follow-up is an untouched multi-speaker test set with blinded listening and replicated trajectories, then randomized host-paired timing with retained raw repetitions, with sparse kernels, quantization-aware training, pinned-calibration INT8, and attention masking in those controls.

6 Conclusion

Under one corpus, one evaluator, and one measurement protocol, no universal winner emerged among the twelve finite-budget states (RQ1): RNDVoC led PESQ and STOI, HiFi-GAN V1 led UTMOS, Vocos led RTF on both lanes, HiFi-GAN V2 led footprint, and RFWave held the intermediate point. Deployment transformations were conditional on architecture, operator graph, and backend (RQ2): FP16 casting was most consistently beneficial, quantization and compilation were heterogeneous, static ONNX INT8 coupled large size reductions to material quality loss, dense masks measured tolerance to zeroed weights rather than sparse execution, and RFWave solver reduction gave the one monotone speed-quality control. At the moderate quality, parameter, and storage vector the artifact-level selector (RQ3) chose FP16-cast Vocos on B200 and dynamic-INT8 Vocos on CPU; the tighter vector selected untransformed Vocos on CPU, and RNDVoC stayed the highest-PESQ feasible state at both vectors, never the RTF-minimizing choice. These conclusions are bounded by proxy metrics, an adaptive single-speaker evaluation set, single trajectories, and non-host-paired timing.

Availability. The source code of both software packages, the complete experimental log, and the admission, exclusion, and result registers under `artifacts/` are public at <https://github.com/rahul1-bot/VOCODE-Efficient-Neural-Vocoders>; the graded state is the latest commit on or before 31 July 2026. LJSpeech is public under its released license, and resolved run configurations record the environment.

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